

The Icosahedral Carrier and the D_4 Lattice

Galois Structure, Reduction Modulo 3 and the Doubling $3 \oplus 3'$ —
an Exact Comparison with $T^4/\mathbb{Z}_3$

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The Icosahedral Carrier and the D_4 Lattice

Abstract

An externally proposed carrier model for emergent physics uses a twelve-port icosahedron with symmetry group A_5 and a local averaging repair rule on its thirty edges. This document compares that carrier exactly with the FFGFT geometry $T^4/\mathbb{Z}_3$ on the D_4 lattice. Four verification scripts establish: (i) the carrier spectrum lies in $\mathbb{Q}(\sqrt{5})$, and the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ exchanges the two three-dimensional A_5 blocks $3 \leftrightarrow 3'$ as spectral projectors, so that over $\mathbb{Q}$ only the 6D block exists – the doubling $6 = 3 \oplus 3'$ of Dok. 285; (ii) the reduction $\mathbb{Z}[\varphi]/(3) = GF(9)$ turns this Galois conjugation into the Frobenius $x \mapsto x^3$ of the FFGFT ground field; (iii) A_5 acts faithfully and absolutely irreducibly on $GF(9)^3$ but not on $GF(9)^2$, hence not in $G(3)$ but in $G(6)$ – the same threshold as the lepton order 26 in Dok. 341; (iv) A_5 is incompatible with the D_4 lattice ($5 \nmid 1152$, Dok. 314); the common core is the tetrahedral group A_4 , which acts simply transitively on the twelve ports, and the port $\mathbb{Z}_3$ is provably never one of the sixteen orbifold generators of $T^4/\mathbb{Z}_3$. The gauge assignment of the external model ($u(1) \oplus su(2) \oplus su(3)$ on $1 \oplus 3 \oplus (3' \oplus 5)$) is representation-theoretically consistent; its Lie structure is an axiom there, not a consequence of the icosahedron.

Status markers

- [B] Algebraically proven (verification script, exact arithmetic).
- [K] Numerically confirmed.
- [S] Structurally plausible, physical identification open.
- [E] Established external mathematics.
- [X] Checked and negative.

1 Orientation: geometry, algebra, and what a field is here

FFGFT has two layers that must not be confused.

The geometry is a torus. The carrier is T^4 with the D_4 lattice (kissing number 24, Dok. 314), and the physics sits in the *modes* of this torus: the winding numbers of the plane waves living on it. The built-in $\mathbb{Z}_3$ symmetry of the lattice turns it into the orbifold $T^4/\mathbb{Z}_3$ with nine fixed points. What a particle is, which mass it carries, which sector it belongs to – these are statements about modes and their $\mathbb{Z}_3$ orbits.

The algebra consists of finite fields: $GF(9)$ as ground field, $GF(27)$ with its 26 units in eight three-orbits, both sitting in parallel inside $GF(729)$ (Dok. 338, 341). The Frobenius $x \mapsto x^3$ is the Galois group of these fields, and the sector pairing $k \leftrightarrow -k$ is its action.

How the layers connect. The fields *are* not the modes. They are the eigenvalue fields, splitting fields and coefficient fields in which the mode operators live, and they decide which modes *can exist*: a mode is an element order in a matrix algebra over $GF(3^k)$, and Lagrange permits only orders dividing the group order $|GF(3^k)^*|$. That is why order 5 exists in $G(3)$ ($5 \mid 80$) while order 26 appears only in $G(6)$ ($13 \nmid 80$, Dok. 341). In short: *the geometry supplies the modes; the Galois structure of the fields says which of them are admissible*. The fields are mode-conditioned in exactly this sense – they are called up by the modes, not the other way round.

Why this organises the comparison. OPH has the same two-layer structure with different entries: the geometry is a sphere of icosahedra, the modes are the blocks of the port space under A_5 , and the algebra is the number field $\mathbb{Q}(\sqrt{5})$ with its Galois group $\sqrt{5} \mapsto -\sqrt{5}$. There too the Galois structure decides which modes exist over the ground field: over $\mathbb{Q}$ only the 6D block, never a single triplet. The comparison in this document is therefore a comparison of the *algebras that classify the modes* – and the two algebras touch exactly once: in the reduction modulo 3, which maps $\mathbb{Q}(\sqrt{5})$ onto $GF(9)$ and the Galois conjugation onto the Frobenius (Theorem 5.2). The geometries, by contrast, touch only in the tetrahedral group and part at the five-fold symmetry (Section 6).

2 Occasion and scope

In September 2026 a carrier model was presented on the IPI list (Observer Patch Holography, hereafter *OPH*) whose hardware is a finite network of twelve-port icosahedra and whose software is a single local rule: on each of the thirty edges (*seams*) the two port readings

are replaced by their arithmetic mean. The author supplied a reproducibility bundle in exact rational arithmetic (300 updates, seeds 7 and 11, an independent verifier).

This document takes no position on the physical claims of OPH. It answers three narrow, exactly checkable questions:

1. What is the structure of the carrier spectrum, and what does its Galois group imply for the distinguishability of the two three-dimensional blocks?
2. How does $\mathbb{Q}(\sqrt{5})$ relate to the FFGFT ground field $GF(9)$?
3. Which part of the icosahedral group A_5 is compatible with the D_4 lattice and the orbifold $T^4/\mathbb{Z}_3$?

The FFGFT reference points are the doubling $6 = 3 \oplus 3'$ (Dok. 285), the embedding $C_3 < A_5$ (Dok. 293), the negative result $5 \nmid |\text{Aut}(D_4)|$ (Dok. 314) and the $G(3)/G(6)$ threshold (Dok. 341).

3 The spectrum of the icosahedral carrier

Theorem 3.1 (Spectrum, [B]). *Let A be the adjacency matrix of the icosahedral graph (12 vertices, 30 edges, 5-regular) and $L = 5I - A$ its Laplacian. Then*

$$\text{spec}(A) = \{5^{(1)}, \sqrt{5}^{(3)}, -1^{(5)}, -\sqrt{5}^{(3)}\}, \quad \text{spec}(L) = \{0^{(1)}, (5 - \sqrt{5})^{(3)}, 6^{(5)}, (5 + \sqrt{5})^{(3)}\}.$$

The only rational non-trivial Laplacian eigenvalue is 6 (multiplicity 5). The field generated by the spectrum is $\mathbb{Q}(\sqrt{5})$, of degree 2.

Proof. Script `spectrum_icosahedral_carrier.py`, SymPy, exact; the seam list is taken from the OPH bundle, alternatively from the canonical vertices $(0, \pm 1, \pm \varphi)$ cyclically. The script also replays the 300 updates of seed 7 in rational arithmetic and reproduces the centered residual recorded in the bundle exactly.

The multiplicities $(1, 3, 5, 3)$ are the dimensions of the irreducible A_5 representations $A \oplus T_1 \oplus H \oplus T_2$ (icosahedral-group notation; H is five-dimensional, T_1, T_2 are the two inequivalent triplets). On a five-fold rotation g_5 they carry the character values

$$\chi_{T_1}(g_5) = \varphi, \quad \chi_{T_2}(g_5) = 1 - \varphi, \quad \chi_{T_1} + \chi_{T_2} = 1.$$

Remark 3.2 (Damping rates, [B]). The expected action of a random single-seam update is $E[M] = I - L/60$ with block eigenvalues $(55 + \sqrt{5})/60 \approx 0.954$ on T_1 , $9/10$ on H and $(55 - \sqrt{5})/60 \approx 0.879$ on T_2 . T_1 is the slowest block; its distance to T_2 is $\sqrt{5}/30$ (ratio ≈ 1.085). This is a statement about rates, not a symmetry selection.

4 Galois conjugation and the doubling $6 = 3 \oplus 3'$

Theorem 4.1 (Galois exchanges $T_1 \leftrightarrow T_2$, [B]). *Let $P_{T_1}, P_{T_2}, P_A, P_H$ be the spectral projectors of L (polynomials in L with coefficients in $\mathbb{Q}(\sqrt{5})$). The Galois automorphism $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ of $\mathbb{Q}(\sqrt{5})$ acts entrywise as*

$$\sigma(P_{T_1}) = P_{T_2}, \quad \sigma(P_{T_2}) = P_{T_1}, \quad \sigma(P_A) = P_A, \quad \sigma(P_H) = P_H.$$

The projector $P_{T_1} + P_{T_2}$ is $\mathbb{Q}$ -rational; P_{T_1} and P_{T_2} separately are not. The minimal polynomial of the T_1 eigenvalue over $\mathbb{Q}$ is $x^2 - 10x + 20$ with roots $5 \mp \sqrt{5}$.

Proof. Script `galois_z3_icosahedral_carrier.py`. The projectors are built as Lagrange polynomials in L ; the exchange is checked by substituting $\sqrt{5} \rightarrow -\sqrt{5}$ and comparing matrices.

Corollary 4.2 (Indivisibility over $\mathbb{Q}$, **[B]**). *Every A_5 -invariant structure on the port space that is defined over $\mathbb{Q}$ as an abstract field – a lattice, an integer character – sees $T_1 \oplus T_2$ as one six-dimensional block; $T_1 \oplus T_2$ is the only six-dimensional submodule with integer character that contains a triplet ($1 \oplus H$ is the other six-dimensional submodule with integer character). Distinguishing T_1 from T_2 is not a symmetry statement.*

Remark 4.3 (The selection happens at the archimedean place, **[B]**). The repair dynamics does not live in the abstract field $\mathbb{Q}$ but in $\mathbb{R}$: $I - L/60$ is a rational symmetric matrix with real eigenvalues, and the two conjugate rates $(55 \pm \sqrt{5})/60$ are the real numbers 0.954 and 0.879. Both embeddings $\mathbb{Q}(\sqrt{5}) \hookrightarrow \mathbb{R}$ yield the same two subspaces; they differ only in which one carries the name T_1 . Iterating $I - L/60$ on any rational vector with a component in the slow block, the ratios of the centered norms – a rational sequence – tend to $((55 + \sqrt{5})/60)^2 = 0.9100$ and not to the conjugate value 0.7733 (script, section 6). Selecting the slow triplet as space is therefore a derivation from the dynamics, as every convergent rational sequence selects the positive root of $x^2 = 2$; it happens at the archimedean place of $\mathbb{Q}(\sqrt{5})$. Geometrically: in the coordinates $(0, \pm 1, \pm \varphi)$ the five seam neighbours of a port sum to $\sqrt{5}v$, the thirty second neighbours (edges of the great icosahedron) to $-\sqrt{5}v$; “ T_1 is space” says that repair couples nearest neighbours.

This is the theorem behind Dok. 285: there the doubling $6 = 3 \oplus 3'$ is the answer to the crystallographic restriction (no icosahedral lattice in three dimensions, but one in six). Here the same statement appears as Galois indivisibility at the *finite* places; the archimedean place separates the triplets, the place 3 does not (Theorem 5.2, Section 8). OPH’s proposal that the three-dimensional continuum is the dense projection of “integer frames built from port labels” is the quasicrystal projection $\mathbb{Z}^6 \rightarrow \mathbb{R}^3$; Dok. 285 describes the cause, OPH the image.

Theorem 4.4 (Restriction to $C_3 < A_5$, **[B]**). *Under a three-fold rotation $g_3 \in A_5$ the port space splits into $\mathbb{Z}_3$ eigenspaces of dimensions $4 + 4 + 4$; blockwise*

$$A \rightarrow 1, \quad T_1 \rightarrow 1 \oplus \omega \oplus \omega^2, \quad T_2 \rightarrow 1 \oplus \omega \oplus \omega^2, \quad H \rightarrow 1 \oplus 2\omega \oplus 2\omega^2.$$

The trivial part has dimension 4, the non-trivial part dimension 8.

This is the C_3 -in- A_5 embedding of Dok. 293 in character form. The non-trivial part (dimension 8) is, as an abstract $\mathbb{Z}_3$ -module, two copies of the orbifold representation on $\mathbb{R}^4$; whether more than representation theory is contained in this is settled negatively in Section 6.

5 Reduction modulo 3: $\mathbb{Z}[\varphi]/(3) = GF(9)$

Theorem 5.1 (Inert prime, **[E]/[B]**). *$x^2 - x - 1$ is irreducible over $GF(3)$ ($5 \equiv 2$ is not a square modulo 3). Hence 3 is inert in $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Z}[\varphi]/(3) \cong GF(9)$.*

Theorem 5.2 (Galois becomes Frobenius, **[B]**). *Under the reduction of Theorem 5.1 the Galois conjugation $\varphi \mapsto 1 - \varphi$ becomes the Frobenius $x \mapsto x^3$ of $GF(9)$: in $GF(9)$ one has exactly $\varphi^3 = 1 - \varphi$.*

Proof. $\varphi^2 = \varphi + 1$, hence $\varphi^3 = 2\varphi + 1 \equiv 1 - \varphi \pmod{3}$. Script `zphi_mod3_a5.py`, sections (1)–(2).

Thus the FFGFT sector pairing (Frobenius on $GF(27)^*$, Dok. 343/363) and the exchange $T_1 \leftrightarrow T_2$ of the icosahedral carrier are not merely analogous: modulo 3 they are the same map.

Theorem 5.3 (A_5 over $GF(9)$, [B]). *The sixty rotation matrices of the icosahedron (entries in $\frac{1}{2}\mathbb{Z}[\varphi]$) reduce modulo 3 to sixty distinct matrices in $GL_3(GF(9))$ spanning all of $M_3(GF(9))$: the reduction is faithful and absolutely irreducible. Entrywise Frobenius of the reduced T_1 matrices yields exactly the reduced T_2 matrices. By contrast $GL_2(GF(9))$ (order 5760, searched exhaustively) contains no $(2, 3, 5)$ -generated subgroup of order 60: A_5 does not embed in $GL_2(GF(9))$.*

Corollary 5.4 (The $G(3)/G(6)$ threshold, [S]). *With $G(3) \cong M_2(GF(9))^2$ and $G(6) \cong M_8(GF(9))$ (Dok. 341), A_5 is not in the unit group of $G(3)$ but is in that of $G(6)$. This is the same threshold at which Dok. 341 finds the lepton order 26. The reasons are formally different (Brauer degrees $\{1, 3, 3, 4\}$ of A_5 in characteristic 3 versus Lagrange in $GF(81)^*$); whether the coincidence has substance remains open.*

6 A_5 versus $\text{Aut}(D_4)$

Theorem 6.1 (Reproduction of Dok. 314, [B]). *The automorphism group of the D_4 root system, recomputed from the 24 roots, has order 1152 with order distribution $(1, 2, 3, 4, 6, 8, 12) \mapsto (1, 139, 80, 228, 464, 144, 96)$ and no element of order 5. Among the 80 elements of order 3 exactly 16 are fixed-point-free on $\mathbb{R}^4$ ($|\det(1 - A)| = 9$); each acts freely on the 24 roots (eight orbits of three).*

Corollary 6.2 ([X]). *A_5 acts on no D_4 lattice: $5 \nmid 1152$. The symmetry group of OPH and the FFGFT lattice structure are incompatible as wholes.*

Theorem 6.3 (The common core is A_4 , [B]). *The elements of A_5 whose matrices in the model $(0, \pm 1, \pm \varphi)$ are integral (no φ) form the tetrahedral group A_4 (orders 1, 2, 3 with counts 1, 3, 8). A_4 acts simply transitively on the twelve ports – the port set is an A_4 -torsor – and embeds as $\text{diag}(M, 1)$ in $W(D_4) < \text{Aut}(D_4)$.*

This sharpens Dok. 285 (" $C_3 < A_5$ "): not only the three-fold symmetry but the whole tetrahedral group of order 12 – the port count – is lattice-compatible; only the five-fold symmetry drops out.

Theorem 6.4 (Ports as half the first shell, [B]). *Every three-fold rotation of A_5 acts freely on the twelve ports (four orbits of three). As a $\mathbb{Z}_3$ -module the port space is $4 \cdot \mathbb{Z}[\mathbb{Z}_3]$, the first D_4 shell is $8 \cdot \mathbb{Z}[\mathbb{Z}_3]$; the ports are one half of " $24 = 12 + 12$ " from Dok. 314.*

Theorem 6.5 (The port $\mathbb{Z}_3$ is not an orbifold $\mathbb{Z}_3$, [X]). *The three-fold rotation permuting the ports is a rotation of $\mathbb{R}^3$ and fixes an axis; as $\text{diag}(M, 1)$ it fixes a line in $\mathbb{R}^4$ and is therefore one of the $80 - 16 = 64$ non-orbifold elements of order 3. This is not an artefact of the embedding: for each of the 16 orbifold generators and each of the 70 choices of four of its eight root orbits (1120 $\mathbb{Z}_3$ -stable twelve-element halves) it was checked whether the twelve roots form a cuboctahedron (D_3 half) or carry the icosahedral graph on some inner-product class. Neither ever occurs.*

Proof. A $\mathbb{Z}_3$ stabilising a D_3 half stabilises its normal up to sign, hence exactly for order 3; it has a fixed vector and is not an orbifold generator. Script `a5_vs_d4.py`, sections (E)–(F).

Remark 6.6 (Side result, [K]). *256 of the 1120 orbifold-stable twelve-element halves are tight frames in $\mathbb{R}^4$ (Gram eigenvalue 6 with multiplicity 4, the 4D counterpart of the OPH relation $G^2 = 4G$ for the port vectors). Their graph spectra contain $\sqrt{13}$, not $\sqrt{5}$; the stabiliser has order 18 and is not transitive. They carry no port structure.*

7 The OPH gauge assignment

OPH assigns to the port module the Lie algebra $u(1) \oplus su(2) \oplus su(3)$: $u(1)$ on A , $su(2)$ on T_1 , $su(3)$ on $T_2 \oplus H$.

Theorem 7.1 (Consistency of the assignment, **[E]**). A_5 sits in $SO(3) \subset SU(3)$ via T_2 . The adjoint of $su(3)$ is $3 \otimes \bar{3} - 1$; under $SO(3)$ it splits as $3 \oplus 5$, under A_5 with the T_2 embedding as $T_2 \oplus H$ ($\Lambda^2 T_2 = T_2$, $\text{Sym}^2 T_2 = 1 \oplus H$). Likewise the adjoint of $su(2)$ under the T_1 embedding is T_1 . The restriction of $u(1) \oplus su(2) \oplus su(3)$ to A_5 is therefore exactly $A \oplus T_1 \oplus T_2 \oplus H$.

Remark 7.2 (What is axiom, **[S]/[E]**). The assignment presupposes two axioms of the external model: (A1) the reversible port response is a compact Lie algebra $\mathfrak{g}$ with a faithful representation on the port module; (A2) A_5 acts on $\mathfrak{g}$ by inner automorphisms. The one-dimensional centre then follows and is not a third axiom: inner automorphisms act trivially on the centre, so it lies in the A_5 -fixed part of $1 \oplus 3 \oplus 3' \oplus 5$, the uniform line. Centre zero would require $\dim \mathfrak{g} = 12$ semisimple, i.e. $su(2)^4$; under an inner action each $su(2)$ factor has fixed dimension 0 or 3, so the fixed dimension would be a multiple of 3, whereas the port module has fixed dimension 1. Hence the centre is the uniform line and $11 = 8 + 3$ is the only decomposition of the rest into dimensions of compact simple algebras. The counterexample $u(1)^3 \oplus su(2)^3$ fails on (A2): under an inner action its module contains no 5; and even without (A2) it fails by Theorem 4.1, since the centre of a torus is defined over $\mathbb{Q}$ and can therefore contain 3 and $3'$ only with equal multiplicity. The correct statement is thus “ A_5 plus (A1), (A2) forces the Lie type”.

FFGFT is in the mirror situation: $su(3)$ follows from the $\mathbb{Z}_3$ triality on T^4 **[B]** (Dok. 321), $su(2)$ is open **[S]** (Dok. 324). In OPH T_1 is clean and $8 = T_2 \oplus H$ is the axiomatic step.

8 Phases as roots of unity: how the connection succeeds

Section 6 and Theorem 5.2 seem to contradict each other: five-fold symmetry is incompatible with the lattice, yet after reduction A_5 sits in $GL_3(GF(9))$. The contradiction dissolves once one distinguishes *as what* a symmetry enters the carrier – as a lattice automorphism or as a phase.

Theorem 8.1 (Phase channel, **[B]**). 1. Modulo 3, φ is a primitive element of $GF(9)^*$: $\varphi^4 = -1$, order 8.
 2. The 72° rotation has characteristic polynomial $(x - 1)(x^2 - (\varphi - 1)x + 1)$. Modulo 3 the quadratic factor has no root in $GF(9)$; its roots lie in $GF(81)$ and have order 5 there ($5 \mid 80$).
 3. $13 \mid 728 = |GF(729)^*|$ but $13 \nmid 80$: order 26 needs the cubic extension (Dok. 341).
 4. In characteristic 3, $x^3 - 1 = (x - 1)^3$: no $GF(3^k)$ contains a primitive cube root of unity. A $\mathbb{Z}_3$ acts unipotently in the field, never as a phase ω .

Proof. Script `phasen_einheitswurzeln.py`.

Reading. A phase is a root of unity, and roots of unity are exactly what a field extension adds. In the lattice the 72° rotation is an automorphism of vectors, and $5 \nmid 1152$. In the field the same rotation is only its eigenvalue ζ_5 , and $5 \mid 80$. Geometric incompatibility becomes arithmetic compatibility as soon as one passes from lattice vectors to phases. This is the mechanism behind Dok. 293: $\theta = 2/9$ arises from R_5 as a phase, not as a lattice operation.

The extension degrees are the two basic operations of FFGFT:

Extension	Degree	carries	Counterpart
$GF(9) \rightarrow GF(81)$	2	$\zeta_5, \varphi, 2/\sqrt{5}$, icosahedron	doubling $6 = 3 \oplus 3'$ (Dok. 285)
$GF(9) \rightarrow GF(729)$	3	ζ_{13} , order 26, leptons	$\mathbb{Z}_3$ triality (Dok. 341)

And three-fold symmetry itself is *not* a phase: in characteristic 3 ω collapses to 1 (Dok. 341: the N eigenvalues $\{-3, +3\}$ become $\{0\}$). This is why in FFGFT the $\mathbb{Z}_3$ is lattice structure – a circulant on the mode orbits, Dok. 314 – and five-fold symmetry is a phase. It is not a choice but forced: what cannot exist as a root of unity in characteristic 3 must enter as a lattice operation; what does not fit the lattice must enter as a root of unity in an extension. The two channels are complementary, and the prime 3 decides which symmetry falls into which channel. In the language of places of $\mathbb{Q}(\sqrt{5})$: the external model chooses between T_1 and T_2 at the archimedean place (Remark 4.3); FFGFT computes at the place 3, where $\sqrt{5}$ has no sign and the two triplets are exchanged by the Frobenius. Both selections are correct, at different places – and that is why the two frameworks touch exactly once. Theorem 6.5 is the geometric side of the same statement: the port $\mathbb{Z}_3$ is lattice, the orbifold $\mathbb{Z}_3$ is lattice, but they are different lattice operations; the five-fold symmetry that distinguishes them is not present in the lattice at all.

9 Summary

	OPH	FFGFT	Link
Field	$\mathbb{Q}(\sqrt{5})$	$GF(9), GF(27) \parallel GF(729)$	$\mathbb{Z}[\varphi]/(3) = GF(9)$; Galois $\rightarrow$ Frobenius [B]
Group	A_5	$\text{Aut}(D_4), \mathbb{Z}_3$	$A_5 \not\subset \text{Aut}(D_4)$; core A_4 (= 12 ports) [B]
$3 \oplus 3'$	T_1 space, T_2 colour	6D lattice (Dok. 285)	indivisible over $\mathbb{Q}$ [B]
$\mathbb{Z}_3$ orbits	4 free on 12	8 free on 24	ports = half the first shell [B]
Port $\mathbb{Z}_3$	axis in $\mathbb{R}^3$	orbifold fixed-point-free	never conjugate [X]
$G(3)/G(6)$	$A_5 \not\subset GL_2(GF(9)), \subset GL_3$	order 26 only in $G(6)$	same threshold [S]

Verification scripts. 2/python/Dok364_Skripte/: spectrum_icosahedral_carrier.py, galois_z3_icosahedral_carrier.py, zphi_mod3_a5.py, a5_vs_d4.py, phasen_einheitswurzeln.py; batch run run_all_364.sh. The seam list of the external bundle is read if present; otherwise the canonical icosahedron is used. No register entry: no earlier document is corrected.